

Yi Lu

Drew University, Department of Mathematics & Computer Science
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EDUCATION

Ph.D. Statistics, The Ohio State University (Columbus, OH), August 2017

- Advisors: Radu Herbei and Sebastian Kurtek
- Dissertation: Function Registration from a Bayesian Perspective

M.S. Statistics, The Ohio State University (Columbus, OH), May 2014

B.S. Mathematics & B.A. History, Mars Hill University (Mars Hill, NC), May 2011

- Honor: Summa Cum Laude

RESEARCH INTERESTS

Functional data analysis, MCMC algorithm, nonparametric Bayesian methods, computational statistics, simulation, curve and image registration, data visualization, applied statistics

PROFESSIONAL EXPERIENCE

Drew University

Associate Professor, Department of Mathematics & Computer Science, Fall 2024 - Present

Norma Gilbert Assistant Professor, Department of Mathematics & Computer Science, Fall 2017 - Summer 2024

Anthology, Inc. (previously iModules Software)

Statistical Consultant, Summer 2019 - Spring 2020

- Project: evaluate existing models and develop new metrics to measure alumni engagement in higher education institutions

The Ohio State University

Graduate Teaching Assistant, Department of Statistics, Fall 2012 - Spring 2017

Consultant, Statistical Consulting Service (SCS), Spring 2016 - Summer 2017

Graduate Research Assistant, Department of Statistics, Summer 2015

- Advisor: Radu Herbei (Supported in part by NSF DMS 1209142)
- Project: compare different methods of parallel computing in executing the Metropolis-Hastings algorithm

The Progressive Corporation (Cleveland, OH)

Research & Development Analyst (internship), Summer 2016

- Project: develop and validate regression models for calculating auto insurance rates

TEACHING

Drew University

(* indicates first offering of the course at Drew; ** indicates a graduate-level or crosslisted graduate level course)

STAT335/DATA503 Applied Regression Analysis:** Fall 2022

STAT350 Statistical Theory: Spring 2018*, Spring 2020, Spring 2022, Spring 2023, Spring 2024

STAT207/STAT117 Introduction to Statistics: Fall 2017, Spring 2018, Summer 2018, Fall 2018, Spring 2019, Summer 2019, Fall 2019, Summer 2020, Spring 2022, Fall 2022, Fall 2023, Spring 2024, Summer 2024, Fall 2024, Spring 2025

DATA200 Data Science: Introduction, History, and Case Studies: Spring 2018*, Spring 2022, Fall 2024

DATA252/DATA551 Modeling and Simulation:** Spring 2020*, Fall 2021, Spring 2023, Spring 2024, Spring 2025

STAT120 Statistical Computing in R: Fall 2019, Summer 2020, Fall 2021, Fall 2022, Fall 2023, Fall 2024

STAT220 Special Topics in Statistics: Bayesian Statistics: Spring 2021*

FIN622 Computational Finance & Large Data Analysis (M.S. of Finance program):** Spring 2019*

DATA251 Data Visualization: Fall 2018*

The Ohio State University

STAT4201 Introduction to Mathematical Statistics: Fall 2015

Recitation instructor, STAT2480 Statistics for Life Sciences: Spring 2015

Recitation instructor, STAT4202 Introduction to Mathematical Statistics II: Fall 2014

Recitation instructor, STAT2450 Introduction to Statistical Analysis: Fall 2014

Recitation instructor, STAT1450 Introduction to the Practice of Statistics: Summer 2014

Recitation instructor, STAT1350 Elementary Statistics: Spring 2014, Spring 2013, Fall 2012

ADVISING

Current students

(* indicates students with an Honors Thesis project)

Arianna Mendez (Drew Summer Science Institute (DSSI), Summer 2024); project: Comparative Analysis of the Effects of Mountaintop Removal Mining on Air and Water Quality In West Virginia

Former students

Muhammad Mir* (co-advisor with Jennifer Olmsted; MATH410: Specialized Honors in Mathematics I, Fall 2023); Honors thesis title: *Optimizing Water Usage and Revenue of Jordan's Agricultural Economy*

Jamie Connors* (STAT300: Independent Study in Statistics, Fall 2022; MATH410: Specialized Honors in Mathematics I, Fall 2022; MATH411: Specialized Honors in Mathematics II, Spring 2023); Honors thesis title: *Two Approaches to Statistical Research: Frequentist and Bayesian* (co-advisor with Sarah Abramowitz)

Alex Goebel (Drew Summer Science Institute (DSSI), Summer 2022; STAT410: Specialized Honors in Statistics I, Fall 2022; STAT300: Independent Study in Statistics, Spring 2023); project: Analyzing Fencing Data

Peter Antico (DATA551: Independent studies in Modeling and Simulation; Spring 2021)

Katelynn Fleming (MATH300: Independent studies in Mathematical Statistics, Spring 2021)

Kevin Lagerquist (MATH300: Independent studies in Mathematical Statistics, Spring 2021)

Member, Honors Thesis Committee

Joseph Rojek, Economics, Spring 2025

Gabriel Dutra, Computer Science and Statistics, Spring 2023

Mehek Agrawal, Biology, Spring 2023

David Nesterov-Rappoport, Computer Science, Spring 2022

PUBLICATIONS

- (5) A. de Voogt, Y. Lu, F. Pan, H. Zuckerman, "Abrupt maneuvers in General Aviation accidents," *Safety Science* (2025): Vol. 185: <https://doi.org/10.1016/j.ssci.2025.106785>.
- (4) Y. Lu, A. de Voogt, F. Gobet, "Dominance, Streakiness, and Non-random Patterns in the Game of N!à," *Journal of Expertise* (2024): Vol. 6(4) & 7(1): https://www.journalofexpertise.org/articles/volume6_issue4_and_volume7_issue1/JoE_6_4_7_1_Lu_et al.html.
- (3) A. de Voogt, A. Campos, Y. Lu "Amphibious Airplane Accidents: An Exploratory Analysis," *Applied Sciences* 13(22): 12224 (2023): <https://doi.org/10.3390/app132212224>.
- (2) Y. Lu, R. Herbei, S. Kurtsek, "Bayesian Function Registration with Random Truncation," *Plos one* 18.7 (2023): e0287734.
- (1) Y. Lu, R. Herbei, S. Kurtsek, "Bayesian Registration of Functions with a Gaussian Process Prior," *Journal of Computational and Graphical Statistics* 26.4 (2017): 894-904.

WORK IN PROGRESS

- (1) M. Mir, Y. Lu, J. Olmsted, "Optimizing Water Usage and Revenue of Jordan's Agricultural Economy"

PRESENTATIONS

Conference Presentations

- (3) *Bayesian Registration of Functions with a Gaussian Process Prior*, Joint Statistical Meetings (JSM), Chicago, August 2016
- (2) *Bayesian Registration of Functions with a Gaussian Process Prior*, International Chinese Statistical Association (ICSA) Applied Statistics Symposium, Atlanta, June 2016
- (1) *Answer in Two Decimal Places Accuracy: How 'Well' Are You Prepared for a Test*, The State of North Carolina Undergraduate Research and Creativity Symposium (SNCURCS), Meredith College, Fall 2010

Invited Talks and Workshops

- (4) *Introduction to Decision Tree & Discussion on Data Analytics*, workshop lecture, Drew Summer College, July 2022
- (3) *Quantitative Graphs: Principles of Effective Data Visualization*, Drew Faculty Workshop Series, May 2022
- (2) *Bayesian Registration of Real-valued Functions*, Seminar Talk, Dr. Bing Zhang Department of Statistics, University of Kentucky, April 2022
- (1) *Bayesian Registration of Functions*, Research Institute for Scientists Emeriti (RISE), Drew University, November 2018

Posters

- (2) *Bayesian Registration of Functions*, Translational Data Analytics Forum, The Ohio State University, October 2015 and November 2016
- (1) *Bayesian Registration of Functions*, Statistics and Biostatistics Graduate Student Poster Session, Department of Statistics, The Ohio State University, September 2015

Miscellaneous

- (3) *The Magic Box: a Rating Engine for Auto Insurance Rate Comparisons* with Stewart Zellars, R&D research meeting, Progressive Cooperation, Cleveland, August 2016
- (2) *A Tutorial on Majorize-Minimization (MM) algorithms* with Sophie Nguyen and Andrew Landgraf, Department of Statistics, The Ohio State University, Fall 2014
- (1) *Temperature Trend of the past 118 Years in the United States* with Po-hsu Chen and Hui Yang, Department of Statistics, The Ohio State University, Fall 2013

GRANTS

- (2) Digital Humanities (DH) mini-grant (\$500) for experimenting classroom technology in DATA251: data visualization, Drew University, Spring 2018
- (1) Digital Humanities (DH) Faculty Development grant for supporting the co-teaching of DATA251: data visualization (with Lee Arnold), Drew University, Spring 2018

AWARDS

- (7) Applied Statistics Symposium Student Travel Award (for contributed paper), International Chinese Statistical Association (ICSA), June 2016
- (6) The Thomas E. & Jean D. Powers Award for Outstanding Teaching Associate, Department of Statistics, The Ohio State University, May 2015
- (5) University Fellowship, The Ohio State University, 2011 - 2012
- (4) Chair's Fellowship, The Ohio State University, March 2011
- (3) Robert W. Sledge Fellowship, Alpha Chi National College Honor Society, April 2011
- (2) Best Undergraduate Paper Award, University of North Carolina at Charlotte, April 2011
- (1) Presidential Scholarship and Grayson Scholarship, Mars Hill University, 2007 - 2011

SERVICES

Department

Coordinator of Statistics Instruction, Spring 2023 - Present

University

Institutional Review Board (IRB), Fall 2021 to present

University Survey Working Group (initialized by the Office of Institutional Research), Fall 2021 to Spring 2023

Global Education Faculty Advisory Committee (GEFAC), Spring 2021 to present

Academic Computing Advisory Committee (ACAC), Fall 2018 to Fall 2020

Statistical Software Discussion Group, Spring 2018

Judge, Fair for Emerging Researchers, March 2018

Professional

Article reviews: Annals of Applied Statistics (June 2022), Bayesian Analysis (May 2022),
Journal of Statistic Education (June 2020, April 2018), Technometrics (October 2019)

PROFESSIONAL MEMBERSHIPS

American Statistical Association (ASA)

International Chinese Statistical Association (ICSA)

MISCELLANEOUS

Computer Skills

R, SPSS, some experience with SAS, C, C++, Java, Stata